TrustVault - Snovasys Requirements & Implementation Report

Snovasys Fresher Hiring Challenge 2026 - RISE (Round 2) | Stack: Java 22, Spring Boot 3.3.4, MySQL 8.0

1. Executive Summary & Challenge Context

TrustVault is an enterprise document security and compliance sharing platform engineered for the Snovasys Fresher Hiring Challenge 2026 (RISE Round 2).

The platform implements strict Zero-Trust Architecture (ZTA), fine-grained Attribute & Role-Based Access Control (ABAC/RBAC), dynamic forensic PDF watermarking, and a cryptographically chained SHA-256 tamper-evident audit ledger.

- **Core Focus:** Preventing Insecure Direct Object References (IDOR), unauthorized file leaks, and database audit tampering.
- **Technology Stack:** Java 22, Spring Boot 3.3.4, Spring Security 6, Spring Data JPA, Hibernate 6.5, MySQL 8.0, Apache Tika 2.9.2, Apache PDFBox 3.0.2.
- **Submission Status:** 100% COMPLETE & FULLY VERIFIED against all mandatory company requirements.

2. Snovasys Company Requirements Verification Matrix

Req ID	Requirement Description	Implementation File(s)	Verification Test	Status
REQ-01	Three-Tier RBAC Architecture (Owner, Viewer,	RoleName.java, SecurityConfig.java	TrustVaultSecurityTests, Live HTTI	COMPLETED AND VERIFIED
REQ-02	Public Owner & Viewer Registration	AuthService.java, login.html, auth.js	AuthControllerIntegrationTest	COMPLETED AND VERIFIED
REQ-03	Public Admin Registration Escalation Preventic	AuthService.java, RegisterRequest.jav	AuthControllerIntegrationTest	COMPLETED AND VERIFIED
REQ-04	Secure Owner Document Upload (10MB, Tika)	DocumentService.java, FileStorageSe	DocumentUploadSecurityTests	COMPLETED AND VERIFIED
REQ-05	Viewer Upload Denial Enforced by Backend	DocumentController.java, SecurityCon	DocumentUploadSecurityTests	COMPLETED AND VERIFIED
REQ-06	IDOR Protection on Direct Document Access	DocumentSecurityService.java	TrustVaultSecurityTests.testIDOR	COMPLETED AND VERIFIED
REQ-07	Access Request & Approval Workflow	AccessRequestService.java, Request(	TrustVaultSecurityTests	COMPLETED AND VERIFIED
REQ-08	Self-Approval Block on Access Requests	AccessRequestService.java	TrustVaultSecurityTests.testSelfAp	COMPLETED AND VERIFIED
REQ-09	Zero-Grace Period Instant Permission Revocat	PermissionService.java, DocumentSe(	TrustVaultSecurityTests.testRevoc	COMPLETED AND VERIFIED
REQ-10	Dynamic Forensic PDF Watermarking	WatermarkService.java, DocumentSer	DocumentUploadSecurityTests	COMPLETED AND VERIFIED
REQ-11	Sequential SHA-256 Audit Trail Chaining	AuditService.java, AuditLogRepository	MySQLDatabaseCrudIntegrationTe	COMPLETED AND VERIFIED
REQ-12	Cryptographic Tamper Verification Engine	TamperCheckController.java, AuditSer	TrustVaultSecurityTests.testTampe	COMPLETED AND VERIFIED
REQ-13	Orphaned Binary Cleanup on DB Transaction F	FileStorageService.java, DocumentSe	DocumentUploadSecurityTests	COMPLETED AND VERIFIED
REQ-14	Production MySQL 8.0 & Standalone H2 Dual I	application-mysql.properties, applicati(	MySQLDatabaseCrudIntegrationTe	COMPLETED AND VERIFIED
REQ-15	1-Click Evaluator Role Switcher & Glassmorph	index.html, login.html, auth.js, dashbo	Live HTTP Probing (Port 8082)	COMPLETED AND VERIFIED

3. Role-Based Capabilities & Boundaries

- **ROLE_OWNER:** Can register publicly, upload documents up to 10MB, view own vault, grant permissions, review and approve/reject viewer requests, revoke permissions instantly, and view audit history.
- **ROLE_VIEWER:** Can register publicly, browse authorized files, download watermarked copies, and submit access requests. Strictly blocked from uploading documents or viewing unshared private files.
- **ROLE_ADMIN:** Must be provisioned through an authorized administrative workflow. Can review full immutable audit ledger, inspect threat metrics, manage user roles, and run cryptographic tamper verification.

4. Security Architecture & Threat Mitigation

TrustVault adopts defense-in-depth at every layer:

1. Magic-Byte Inspection: Apache Tika inspects binary stream magic numbers to prevent disguised web shells (e.g. .exe or .sh renamed to .pdf).

2. Non-Repudiation Watermarking: Every viewer download dynamically renders viewer username, client IP, and UTC timestamp diagonally across all pages via Apache PDFBox.

3. Merkle-Style Hash Pointers: Each audit record cryptographically binds the current action with the hash of the preceding block. Any tampering or deletion breaks the hash sequence and pinpoints the exact compromised record.